FineTuneIT Service: Making Fine-Tuning Simple

IISc - AI & ML Ops Cohort 4 Group 3

Agenda

- Motivation
- Technical Architecture & Innovation
- FineTuneIt Platform: Core Components, Tech Stack & System HLD
- Model Portfolio & Optimization Strategy
- Experimental Validation & Results
- Platform Workflow & User Experience
- Industry Alignment & Impact
- Competitive Advantages & Future Roadmap
- Appendix

Motivation

Motivation: Why FineTuneIT?

- **Our Vision:** To create an unified **optimized platform** that delivers a **seamless finetuning service experience .**

Our Solution: FineTuneIT Service

- **GPU On-Demand:** Access computing power precisely when needed, eliminating idle costs and provisioning delays.
- **Diverse Base Model Options:** A wide array of pre-trained base models available on a single platform, simplifying model selection.
- **Seamless Experience:** Designed for ease of use, enabling efficient and effective finetuning for all users.

Key Engineering Differentiators

- **Distributed & Scalable Pipelines:** Robust infrastructure for high-performance, scalable finetuning workflows.
    - Intelligent pipeline with diverse base model selections for optimal results.
- **Specialized for text-only Tasks:** Engineered to excel in finetuning for complex reasoning and non reasoning tasks

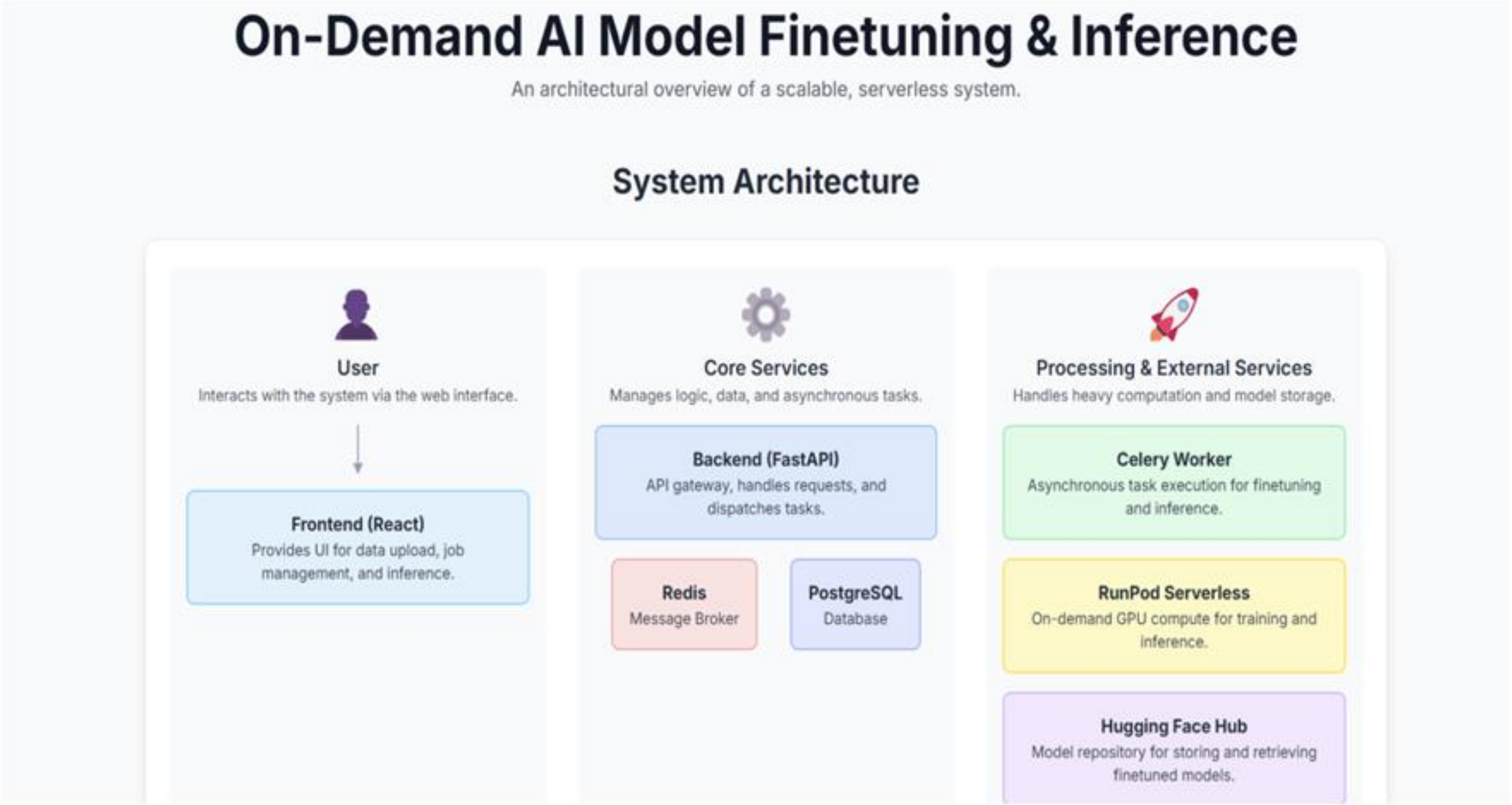
Technical Architecture & Innovation

Core Technical Innovations

- **Unsloth Framework Integration:**
  - 2x-30x faster training speeds
  - 70-90% VRAM reduction
  - Zero accuracy loss
- **RunPod Infrastructure:**
  - 70% cost reduction vs. major clouds
  - Serverless GPU scaling (A100/H100)
  - Optimized for AI workloads

Result: Production-grade fine-tuning accessible through a web interface

FineTuneIt Platform: Core Components



FineTuneIt Platform: Tech Stack

Technology Stack

Frontend

React.js: A powerful JavaScript library for building dynamic and interactive user interfaces.

Backend

FastAPI: A modern, fast web framework for building APIs with Python, featuring automatic documentation.

Database

PostgreSQL: A robust, open-source object-relational database system used for storing job metadata and results.

Task Queue

Celery & Redis: Celery is a distributed task queue that handles long-running processes asynchronously. Redis acts as the message broker, facilitating communication between the web server and the workers.

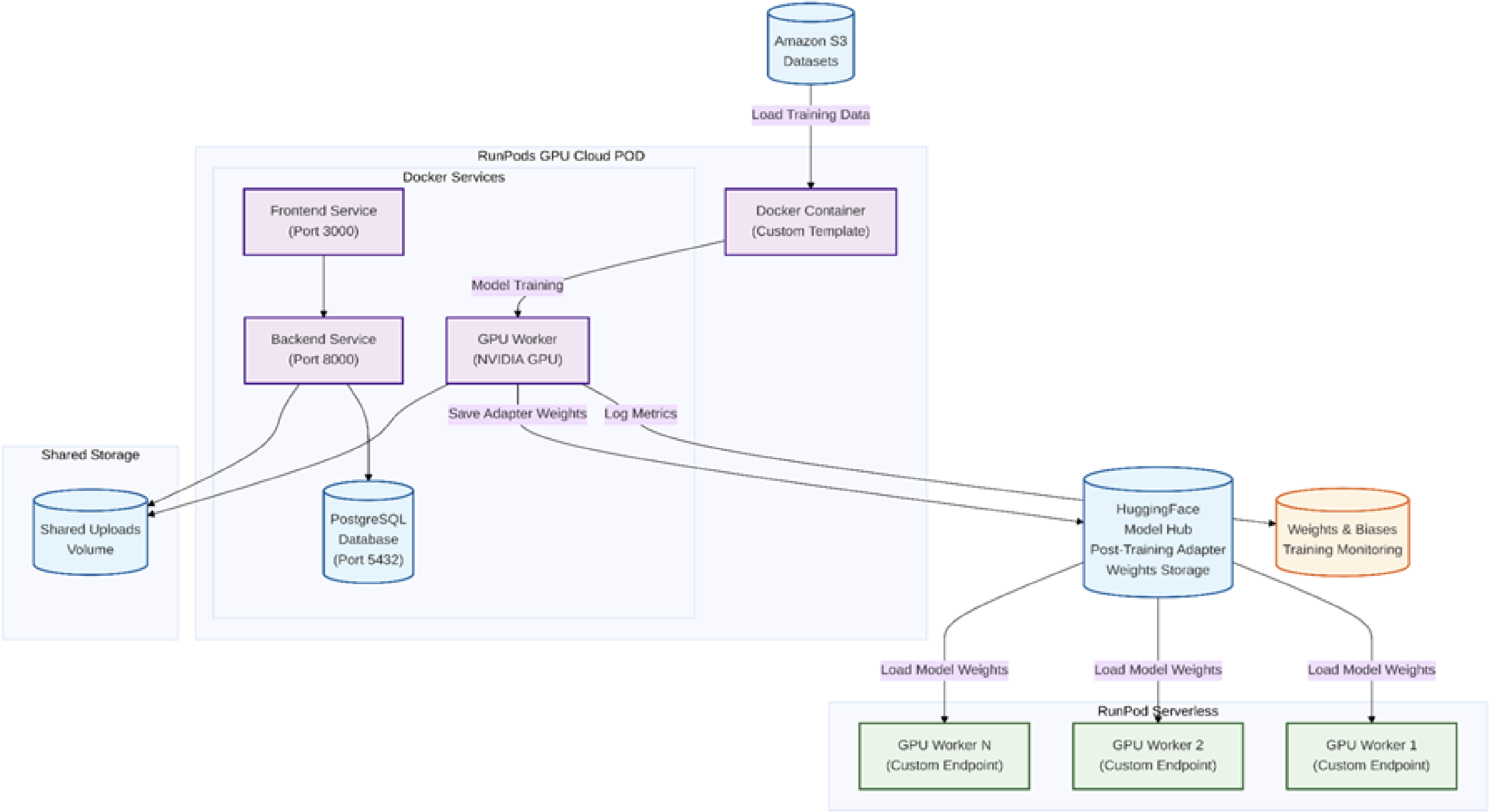
Compute & Optimization

RunPod Serverless: Provides on-demand, pay-per-second GPU infrastructure, eliminating the need to manage servers. **Unsloth:** A library for faster, more memory-efficient finetuning of LLMs, reducing compute costs.

Model Hub

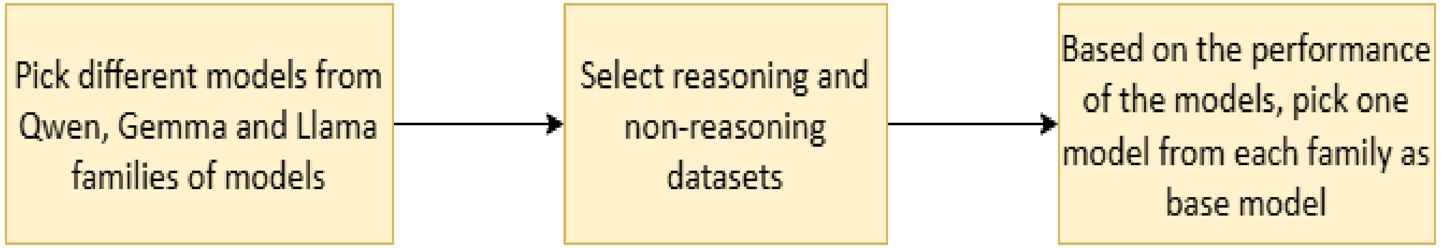
Hugging Face Hub: A platform for hosting and sharing machine learning models, used here to store and version control the finetuned models.

FineTuneIt Platform: High Level Design



Model Portfolio and Optimization Strategy

The aim of the experimentation is to select the best general-purpose base models to be provided on the platform for users to select based on specific use-cases.
4-bit quantized versions of the models have been used from Unsloth

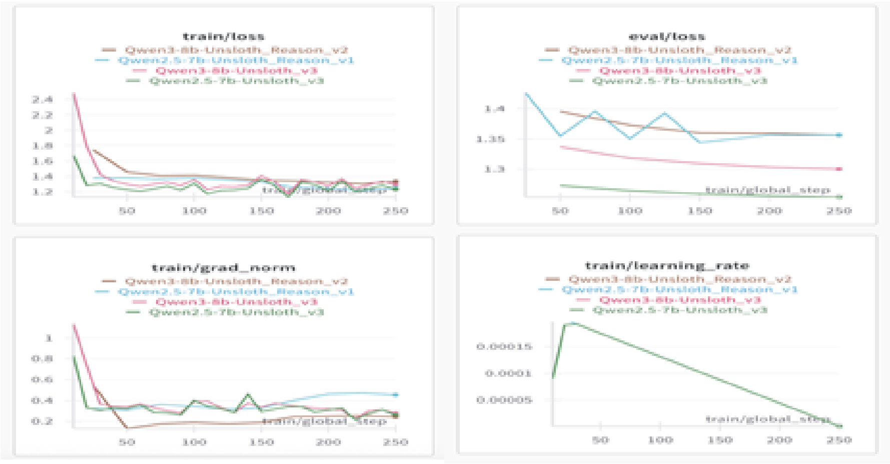


Model	No. of parameters	Recommended for	Other versions tried	Purpose of selection over other version
Qwen 2.5	7B	Reasoning tasks (e.g., medical, logical)	Qwen3-8B	Qwen3 has frequent runtime instability (e.g. PassManager::run failed errors)
Gemma	7B	Non-reasoning tasks and on-device or limited infra setups	Gemma 2B	Gemma 2B shows significantly weaker performance on reasoning models
Llama 3	8B	Mixed reasoning and generation use cases	-	Consistently strong across reasoning and non-reasoning tasks

Experimental Validation and Results

For the purpose of experimentation, the following observations have been made (graphs in appendix):

	Non-reasoning dataset		Reasoning dataset	
Model	**Training stability**	**Eval loss**	**Training stability**	**Eval loss**
Qwen2.5-7B	Stable	Low (converges well)	Stable	Low (but higher than non-reasoning)
Gemma 7B	Stable	Low (converges well)	Stable	Moderate but consistent
Llama3-8B	Stable but needs good batch tuning	Low (converges well)	Stable	Low



Platform Workflow & User Experience

Fine-Tuning Pipeline (Automated)

Data Upload: HuggingFace format validation → PostgreSQL job creation

GPU Processing: Serverless RunPod execution with Unsloth optimization

Model Deployment: Automatic HuggingFace Hub integration

Monitoring: Real-time W&B tracking with comprehensive metrics

User Experience Design

Zero-Code Interface: Web-based platform requiring no ML expertise

Real-time Feedback: Live training progress and performance monitoring

One-Click Deployment: From dataset to deployed model in hours

Industry Applications & Impact

Cross-Industry Deployment (*non-exhaustive)

- **Healthcare:** Clinical decision support, medical text analysis
- **Legal:** Contract analysis, document processing, compliance automation
- **Financial:** Risk assessment, report analysis, customer support
- **Enterprise:** Knowledge assistants, content generation, internal documentation

Business Impact

- **Time Reduction:** Weeks → Hours for model deployment
- **Cost Savings:** 70% reduction in infrastructure costs
- **Accessibility:** Non-technical domain experts can fine-tune models
- **Quality Maintenance:** Production-grade performance with automated optimization

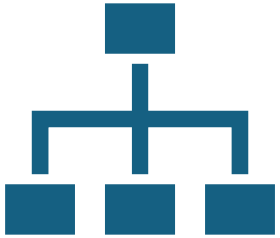
Competitive Advantages & Future Roadmap

Technical Differentiators

- **Unified Platform:** End-to-end solution vs. fragmented tools
- **Performance Leadership:** Unsloth integration for superior efficiency
- **Cost Optimization:** Serverless architecture with intelligent resource allocation
- **Enterprise Readiness:** Monitoring, scalability & more

Future Roadmap

- **RL Based Agent Fine-Tuning** to perform workflow based tasks
- **Multi-modal support** (vision/audio), distributed training
- Federated learning, custom architectures, enterprise integrations

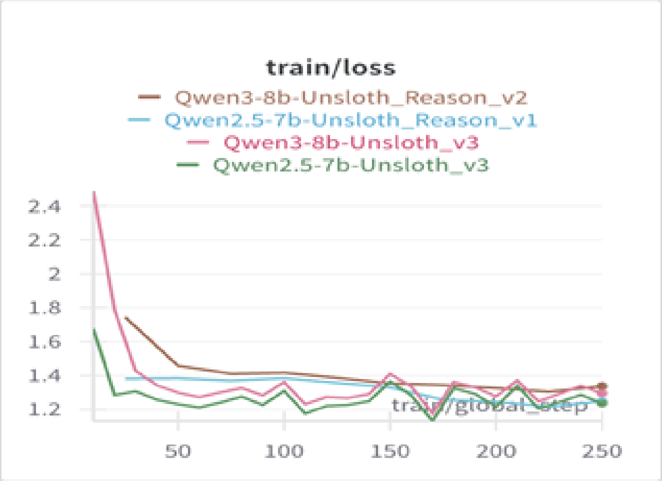


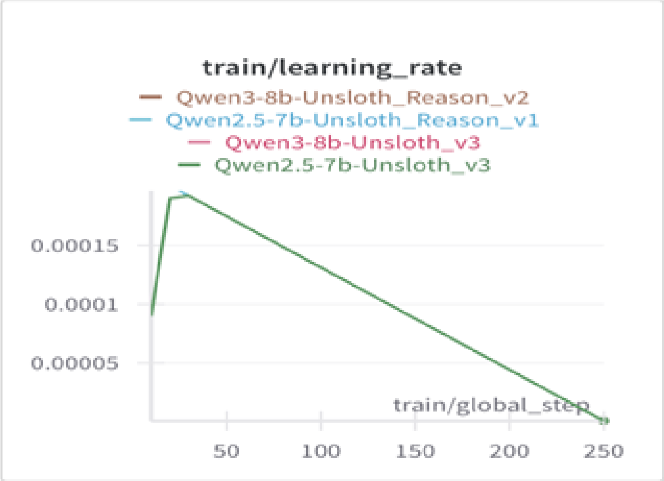
Vision: State of the Art, Cost-efficient and Easy to use Platform for domain and task specific AI adaptation for B2B as well as B2C Buyer/ User Personas

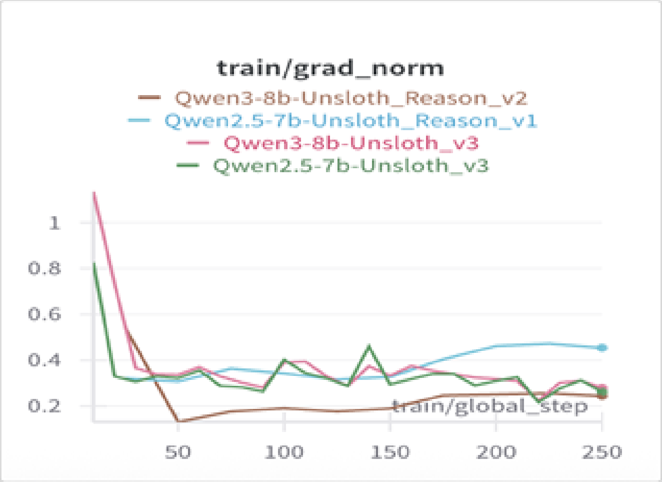
🌍 Democratizing custom LLMs for enterprises & researchers.

💡 Abstracts ML complexity while preserving power & flexibility.

Appendix: Experimental Validation & Results



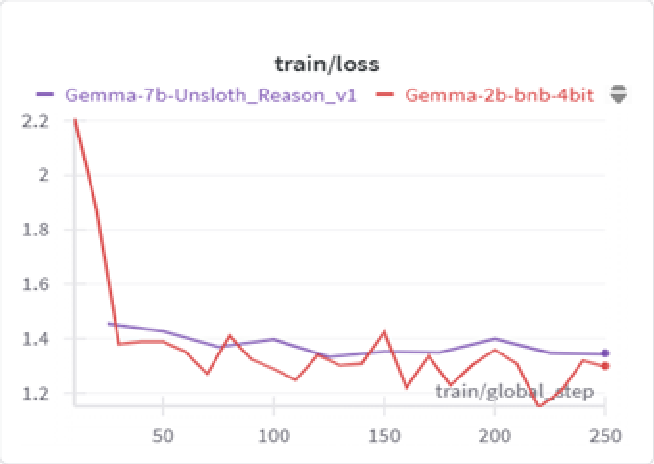


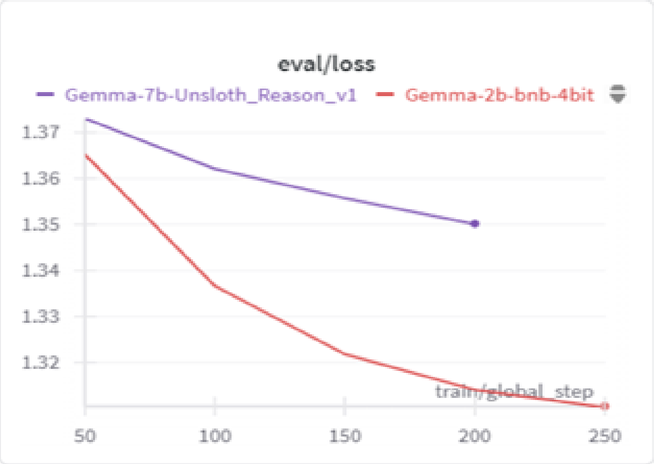


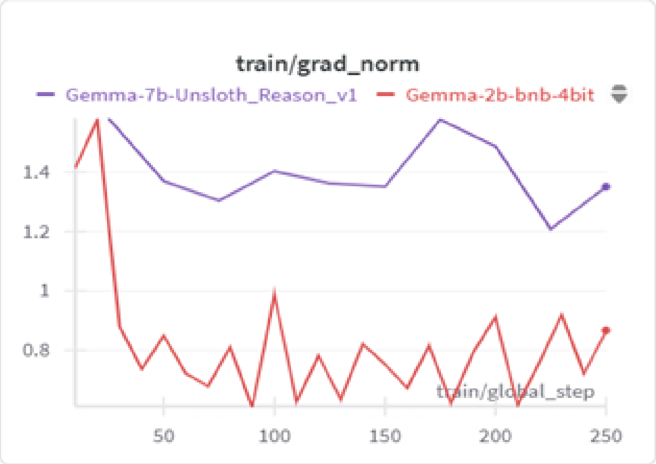
Insights:

- Decreasing curve for eval loss
- Training stability observed
- Qwen2.5-7B has the lowest eval loss for non-reasoning dataset
- The loss curves Qwen2.5-7B and Qwen3-8B overlap with increase in the steps

Appendix: Experimental Validation & Results







Insights:

- Decreasing curve for eval loss
- Training stability observed in 7B model for reasoning

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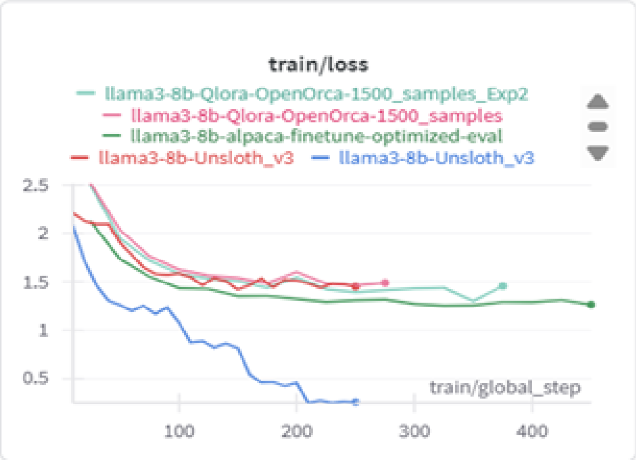

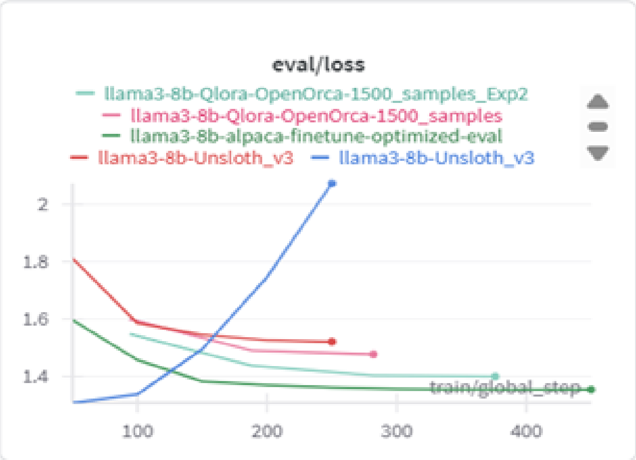

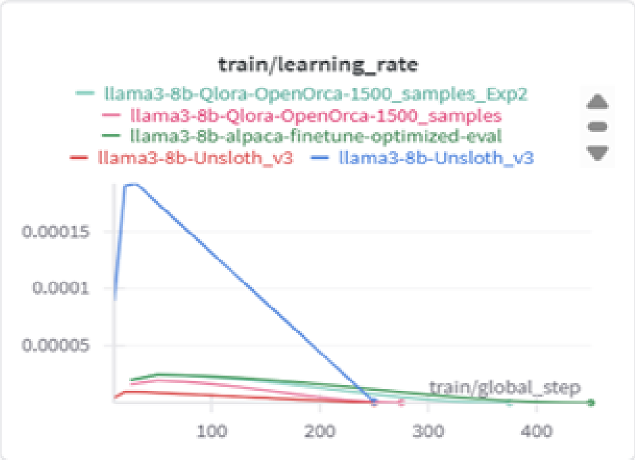

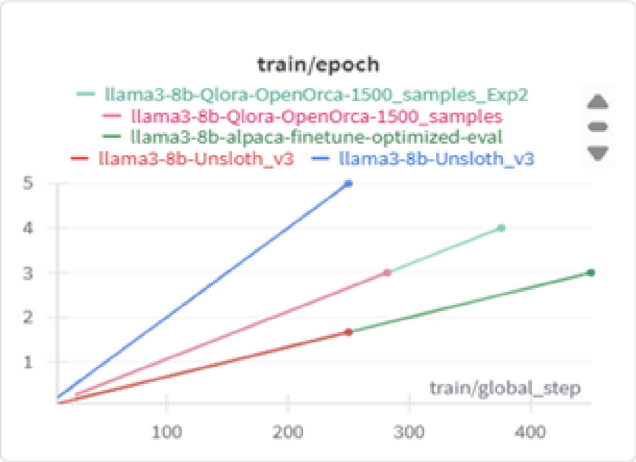

Insights:

- Stable training curve
- Eval loss decreases for all iterations except llama-3-8b

Thank you